

A Tutorial on the Ex-Ante ERW Project

What the model does, in plain language

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2026-05-26

Abstract

This is a short tutorial that walks through what the GAIA *Ex-Ante ERW* project does and why. It is written for someone who has not seen this work before — no background in soil chemistry, carbon markets, or geospatial modelling is assumed. Every technical term is defined on first use, and the glossary at the back gives a one-line refresher for each one. Read it once, look at the figures, and you should know enough to follow any conversation about the project.

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1. What is this project, in one paragraph?

We are building a **map of where in Sub-Saharan Africa it would make economic sense to use crushed rock to (a) reduce soil acidity for farmers, and (b) pull carbon dioxide out of the atmosphere at the same time**. The map answers, for every roughly-1 km² patch of cropland on the continent, the question: *if a farmer spread basalt rock dust on this field, would the carbon-removal payments plus the yield improvement outweigh the cost of buying, transporting, and spreading the rock?* The project produces this map under different policy assumptions (the carbon price, the cost of measurement and verification, the choice of rock, how finely the rock is ground), and ranks the locations that look the most promising first.

2. Why this matters

The world has agreed it needs to *remove* carbon dioxide from the atmosphere, not just stop emitting more, if it wants to hold warming below 2 °C — this is the conclusion of the 2023 IPCC Sixth Assessment Synthesis Report [1]. **Enhanced rock weathering** — the practice of spreading finely-

ground silicate rock (almost always basalt) on cropland — is one of the few removal pathways that

1. is **physically scalable** — Beerling and colleagues [2] estimate that deploying basalt across global croplands could remove 0.5 to 2 billion tonnes of CO₂ per year by 2050, with the highest per-hectare potential in the warm-humid tropics,
2. has a **co-benefit for farmers** (it works like agricultural lime, reducing soil acidity and raising yields) — a US Corn-Belt field trial reported a cumulative 10.5 ± 3.8 tonnes of CO₂ removed per hectare and a 12–16 % yield uplift over four years [5]; a 2024–25 smallholder trial in Kisumu County, Kenya reported a 71 % first-year and 79 % second-year maize-yield increase from a single 20 t/ha nephelinite (volcanic rock powder) application [6],
3. uses **existing infrastructure** (rock from quarries that already exist; tractors that already spread lime; logistics chains that already exist), and
4. has **measurable, verifiable** outcomes that carbon-credit buyers are willing to pay for.

The headline numbers are not uniformly rosy: a recent three-year Swiss vineyard trial measured only about 100 ± 30 kg CO₂ per hectare per year [7], an order of magnitude below the higher published rates. This underlines that *climate, soil pH, and drainage matter enormously*. The high-end ERW response is concentrated in warm, mildly-acidic, well-drained, vegetated systems — exactly the agroecological space most of cropland Sub-Saharan Africa occupies.

Sub-Saharan Africa is a particularly interesting place to apply this because (i) its soils are widely acidic and lime-poor, so the agronomic benefit is unusually large; (ii) its climate is warm and humid, which is exactly the climate that makes the rock dissolve quickly and remove the most carbon; and (iii) its farmers are largely cut off from carbon markets, so an enhanced-rock-weathering programme would deliver a new income stream rather than crowding out an existing one.

But for a policy-maker or an investor, the right question is more specific than “is this a good idea in general?” The right question is: **at what carbon price, in which country, on which crop, would this actually be profitable today?** That is the question the project model answers, location by location.

3. What does the model actually compute?

For every 1 km² patch of cropland in 47 Sub-Saharan African countries, the model computes a single number: the **per-hectare gross margin** of a basalt-application programme on that patch. In plain English, the number is

*what the farmer earns from the carbon-credit payment, **plus** what they earn from the bigger harvest, **minus** what it costs to buy, grind, transport, and spread the rock.*

That number can be positive (profitable to do here), negative (unprofitable), or close to zero (could go either way depending on assumptions). When we put all the patches on a map, we see immediately where the deployment-friendly zones are.

4. The conceptual picture

The conceptual framework, summarised in **Figure 1**, is the single most-important diagram in the project. It shows that the profitability of a patch of cropland comes from **three independent chains of cause and effect**, and that those chains converge on the single profitability number described above.

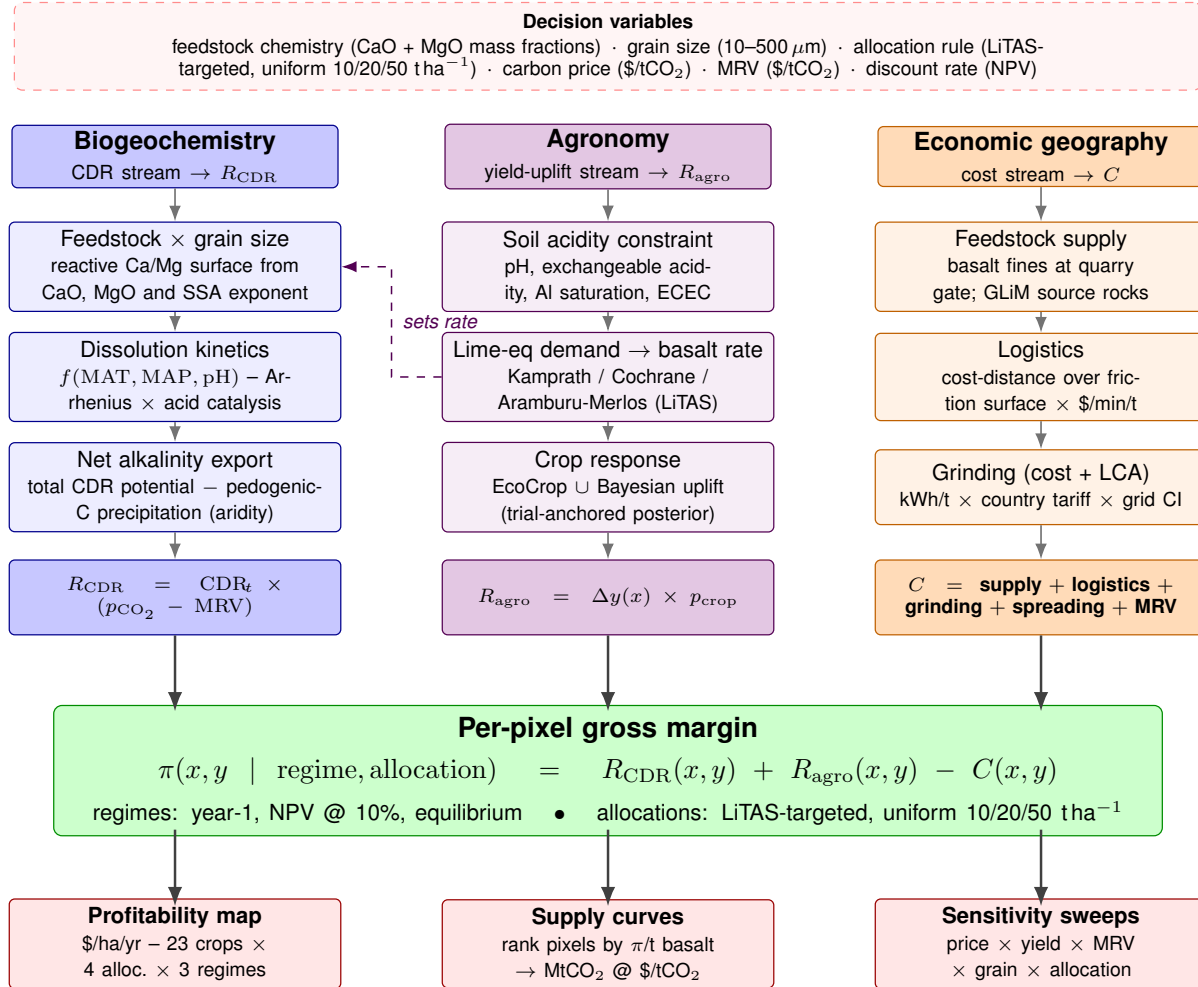


Figure 1: Figure 1. The conceptual framework. Three streams (left to right: *biogeochemistry* $\square$ carbon revenue; *agronomy* $\square$ yield revenue; *economic geography* $\square$ total cost) converge on a single per-pixel gross-margin equation. The six dashed boxes at the top are the user-settable decision dials. The three coloured boxes at the bottom are the three outputs the model produces. The dashed arrow from agronomy back to biogeochemistry captures the fact that the same basalt application rate that the farmer needs for soil-acidity reasons is the basalt that the biogeochemical chain dissolves to remove CO₂.

The three chains are:

4.1 The biogeochemistry chain (carbon side, blue in Figure 1)

This is the chain that tells us **how much CO₂ a tonne of basalt will actually remove** when it sits on a particular farmer's field for a year. The chemistry, summarised by Hartmann and colleagues

[4], says that when basalt dissolves in moist soil, the calcium and magnesium in the rock combine with atmospheric CO₂ to form *bicarbonate*, which then washes into rivers and eventually to the ocean, where it stays for tens of thousands of years.

Three things determine how much of this actually happens on a given field:

1. **What's in the rock** (its calcium and magnesium content, default 10 % CaO and 7 % MgO by mass, and how finely it is ground — a finer grind exposes more surface to the soil water and dissolves faster [3, 11]).
2. **The climate** (the rock dissolves faster in warm, wet conditions — our reference point is a temperate American climate of 11 °C mean annual temperature and 1000 mm mean annual precipitation, anchored to the Lewis et al. [11] basalt-weathering calibration; tropical sites typically dissolve 1.5 to 2 times faster).
3. **The local soil pH** (the more acidic the soil, the faster the rock dissolves — this is the chemistry of acid-catalysed weathering, peaking near pH 5).

We also subtract a **pedogenic-carbonate** deduction — a fancy name for the well-known fact that in dry regions, some of the alkalinity the rock releases doesn't make it to the ocean. It precipitates as solid carbonate right there in the soil, which doesn't count as a permanent carbon removal. We use the aridity bands keyed on mean annual precipitation from the UNEP World Atlas of Desertification [12]; this deduction can take away as much as 90 % of the gross removal in arid pixels.

The output of this chain is the **CDR yield surface**: for every 1 km² patch, the tonnes of CO₂ that one hectare of treated cropland would remove, accounting for climate, pH, and aridity penalties.

4.2 The agronomy chain (yield side, purple in Figure 1)

This chain tells us **how much extra crop the farmer will harvest** if they apply basalt. The mechanism is almost the same as agricultural lime: the dissolved cations from the rock raise the soil pH and free up nutrients, particularly phosphorus and the basic cations themselves.

Three things determine the yield gain:

1. **How acidic the starting soil is** — the soil pH, exchangeable acidity, and aluminium saturation rasters come from SoilGrids-Africa [13]. On a soil already at pH 7, basalt does very little for the yield; on a soil at pH 5.0, it can do a great deal.
2. **The crop being grown** — maize is moderately acid-sensitive, so it responds strongly; cassava is acid-tolerant, so it responds less. We have 23 crops mapped across Africa using the SPAM database [17].
3. **How much rock you apply** — we use the `limer` R package [14] to compute the *lime-equivalent* requirement using the LiTAS recipe of Aramburu Merlos et al. [15], then convert lime-equivalent demand to a basalt application rate using the rock's chemistry. That same rate is what feeds into the biogeochemistry chain (this is the dashed cross-arrow in Figure 1).

The yield gain is then multiplied by the local crop price (taken from the United Nations FAOSTAT producer-price database, 2016–2020 average) to give the **yield revenue surface**: dollars per hectare per year of agronomic benefit.

4.3 The economic-geography chain (cost side, orange in Figure 1)

This chain tells us **how much it actually costs** to put a tonne of basalt on a farmer's field. The cost has four components:

1. **Buying the rock from the quarry** — typically \$10 per tonne, because basalt fines (the pieces too small to use as construction aggregate) are a by-product that quarries are happy to sell cheap. We use the Global Lithological Map of Hartmann and Moosdorf [20] to find where basalt outcrops actually are.
2. **Transporting the rock from the quarry to the farm** — this is by far the most variable component. A farmer 50 km from a basalt quarry pays a few dollars; a farmer 500 km from one pays much more. We calculate this using the *cost-distance* over a 1-km global friction surface from the Malaria Atlas Project [19], which knows the locations of roads, rivers, and rough terrain. The default haulage cost is \$0.04 per minute per tonne (a 30-tonne truck at \$80 per hour); we cap the haul at 1000 km because beyond that the diesel lifecycle emissions of the truck erase the CDR credit.
3. **Grinding the rock fine enough to dissolve** — this is an electricity cost, so it varies by country (industrial electricity tariffs in Sub-Saharan Africa range roughly \$0.05 to \$0.20 per kWh).
4. **Spreading the rock on the field** (\$8 per tonne) and **measuring/verifying the CDR for the carbon market** (\$20 per tonne of CO₂ removed under the 2026-05-26-refreshed defaults — midpoint of Levy et al. 2024 first-of-kind pilots and the at-scale aggregator-pooled forecast).

The output of this chain is the **delivered cost surface**: dollars per tonne of basalt that arrives at a farmer's field.

4.4 Where the three chains meet

The three chains all produce numbers in the same units (US dollars per hectare per year, on the same 1 km² grid). The model simply adds the two revenue streams and subtracts the cost stream to get a single profitability map. That is equation (1) in the working paper; for this tutorial, the words above are the equation.

4.5 The six “dials” at the top of Figure 1

The boxed items at the top of Figure 1 are the **decision variables** — the things the user can change before running the model. They are:

1. **The chemistry of the rock** (a richer-in-calcium rock removes more carbon per tonne).
2. **How finely the rock is ground** (a finer grind dissolves faster but costs more electricity to produce).

3. **Where on the landscape we apply it** (“targeted” — only on farms where the soil acidity is high enough to need it; or “uniform” — everywhere, regardless of acidity).
4. **The carbon price** in dollars per tonne CO₂.
5. **The MRV cost** (measurement, reporting, verification — what we pay to convince the carbon market that the removal actually happened).
6. **The discount rate** (how much we prefer dollars now over dollars in five years).

By running the model with different settings of these dials, we can produce supply curves and sensitivity charts (the two outputs on the right of Figure 1) that tell us, for example, “at \$80 per tonne of CO₂ and a 200 µm grind, the model predicts 12 million hectares are deployable”.

5. The modelling tasks, in plain language

The model is built as a pipeline of **nine functional blocks**, each doing one well-defined job. Figure 2 shows how they connect. Below I walk through each one in plain English.

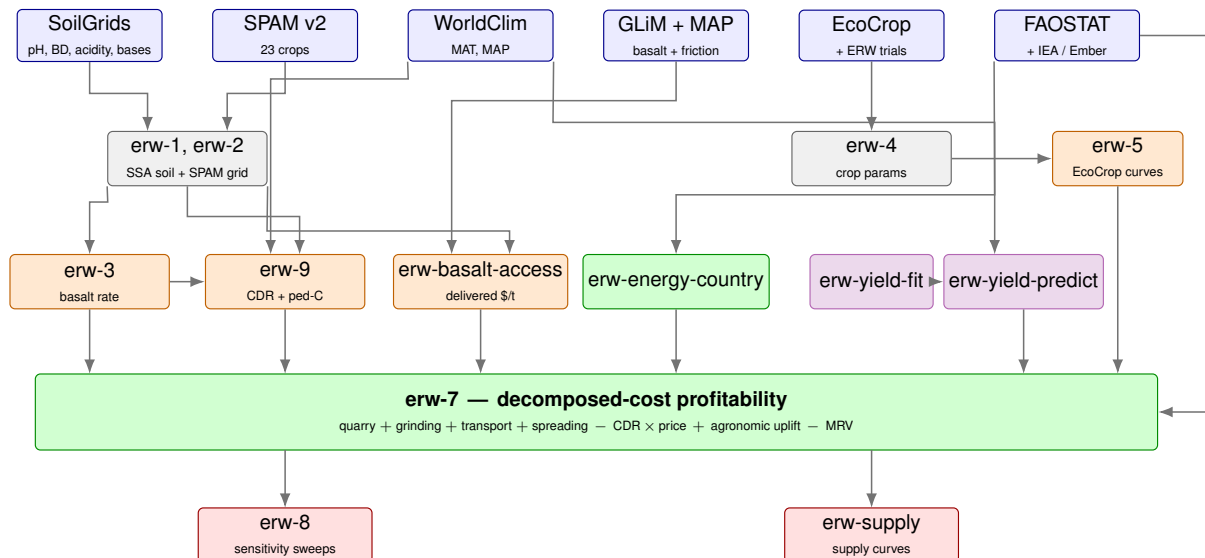


Figure 2: Figure 2. The pipeline architecture. Blue boxes are raw data sources; grey boxes are the working-grid step; orange boxes are geospatial processing; green boxes are economic; purple boxes are the Bayesian-statistics step; the wide green bar in the middle is the profitability function; the red boxes at the bottom are the deliverables. Every arrow is a data dependency: the box at the tail of the arrow produces something that the box at the head of the arrow consumes.

Block 1 — Build the working map

We take 47 country boundaries from GADM v4, the SoilGrids-Africa database [13] (which gives us soil pH, density, and chemistry at every kilometre on the continent), and the MapSPAM crop-production database [17] (which tells us where which of 23 crops is actually being grown in Africa). We put all of these on a common grid at roughly 9 km resolution. Every subsequent step operates on this same grid.

Block 2 — Figure out how much rock each pixel needs

For every patch of cropland, we use the standard agricultural lime-requirement formulas (the same ones a soil-testing lab would use), implemented in the `limer` R package [14], with the LiTAS recipe of Aramburu Merlos et al. [15] as the default. We then convert that lime requirement to a basalt requirement, using the rock's chemistry (default 10 % CaO + 7 % MgO by mass, with a Lewis-2021 [11] reactive-fraction calibration of 0.20 at the reference grain × climate). This gives us a **basalt application rate** in tonnes per hectare for every patch.

Block 3 — Compute the CDR yield

For every patch, we take the per-tonne carbon-removal potential of the chosen rock (computed from its calcium and magnesium content), multiply it by the climate factor (warmer-and-wetter sites dissolve the rock faster), multiply by the pH factor (acidic soils dissolve the rock faster), and subtract the pedogenic-carbonate fraction (the fraction of alkalinity that gets locked up locally and never makes it to the ocean). The result is the **tonnes of CO₂ removed per hectare per year** for that patch, under our default assumptions.

Block 4 — Compute the delivered rock price

This is the trickiest geospatial step. For every patch of cropland, we ask: *what is the nearest basalt outcrop, and how many minutes of truck-time away is it?* We use the Global Lithological Map [20] for the location of basalt outcrops and the Malaria Atlas Project 2019 motorised friction surface [19] for the per-pixel travel speed (it accounts for roads, terrain, water bodies). The cost-distance is accumulated using terra's algorithm; multiplying by the hourly cost of a truck (\$80 per hour for a 30-tonne diesel truck) gives us the **delivered price** of a tonne of basalt at every cropland pixel.

Block 5 — Country-resolved electricity

For every country in Sub-Saharan Africa, we have a tabulated industrial electricity price (sourced from IEA, Ember, and GET.invest 2022–2024 data, so we know what it would cost to grind a tonne of rock there) and a tabulated grid carbon intensity (so we know how much CO₂ the grinding emits, which we have to subtract from the gross CDR to be honest about the net climate benefit).

Block 6 — Crop response

The trickiest *biological* step. We have two ways to predict how much extra crop a farmer will get from a basalt application. The first is **EcoCrop** — a long-established suitability model that gives a 0-1 score for every crop on every soil, available in R as the `Recocrop` package by Hijmans and Manners [16]. The second is a **Bayesian hierarchical statistical model**, implemented in `brms` [18] and fitted to a small set of published ERW field trials (currently 14 rows). The Bayesian model is more flexible because it lets us update our predictions as new field trials are published. The profitability block uses the Bayesian model when it has run; otherwise it falls back on EcoCrop.

Block 7 — The profitability function

This is the block that brings together everything above. It applies the gross-margin equation to every patch of cropland, for every crop, for every choice of *how much rock* (targeted vs uniform 10, 20, 50 tonnes per hectare), and for every choice of *time horizon* (year-1, ten-year net-present value at 10 % discount, or equilibrium). The output is a large stack of map layers — 276 of them per run: 23 crops × 4 allocation rules × 3 regimes. Default policy parameters: carbon price \$150/tCO₂ (the 2024–25 mid-market for verified ERW credits), MRV cost \$20/tCO₂, grain size 50 µm, discount rate 10 % per year, CDR phasing 30/25/20/15/10 % over five years.

Block 8 — The supply ranking

In practice, basalt supply is constrained. We can't quarry an infinite amount of basalt overnight, and we can't grind it instantly. So the supply block ranks every patch of cropland from “best dollar of value per tonne of basalt” to “worst dollar of value per tonne of basalt”, and produces a **supply curve**: how many tonnes of CO₂ would we remove each year if our annual basalt supply were 1, 5, 10, 50, 250, or 500 million tonnes? This is the headline answer for an investor: *for \$X per year of basalt supply, you get Y million tonnes of CO₂ per year removed at a marginal price of \$Z per tonne.*

Block 9 — Sensitivity sweeps

Finally, we run the whole pipeline five times under different assumptions, to see which of our assumptions matter most for the answer. The five sweeps are: across different crop, basalt, and carbon prices; across different yield-uplift levels; across different MRV costs; across different grain sizes; and across different allocation rules.

6. What the maps look like

To make the model concrete, we have rendered twelve “indicator maps” that show the inputs and intermediate outputs across Sub-Saharan Africa. Below are three of the most diagnostic ones.

7. What we have actually produced so far

By **2026-05-26**, the model is in the following state:

- All five of the input-and-surface blocks (1, 2, 3, 4, 5) have been run end-to-end on real data and produce the maps above.
- The pH map matches the qualitative geography of African soils that you would expect from any soil-science textbook.
- The CDR yield map peaks in the humid tropics, exactly where the literature [2, 5] says it should. With the 2026-05-26 round-1 refresh (50 µm grain, CGIAR-CSI AI for the pedogenic deduction), mean CDR under LiTAS-targeted allocation lands at **3.5 t CO₂/ha** cumulative (was ~2 t/ha) and the per-tonne CDR potential at the reference climate is **88 kg CO₂/t basalt** (was 62 kg/t).

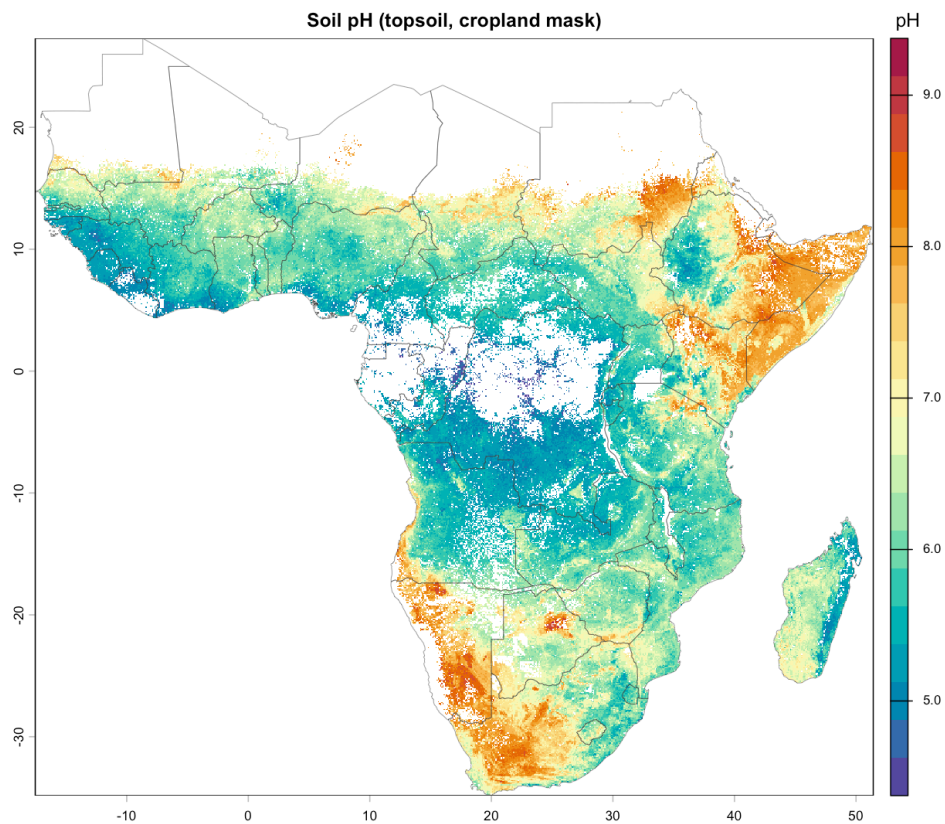


Figure 3: Figure 3. Soil pH across Sub-Saharan African cropland. The humid-tropical zones (West Africa, the Congo basin, East Africa) have acidic soils (orange-to-red); the Sahel and parts of southern Africa have alkaline soils (blue). The acidic zones are where the agronomic benefit of basalt is large.

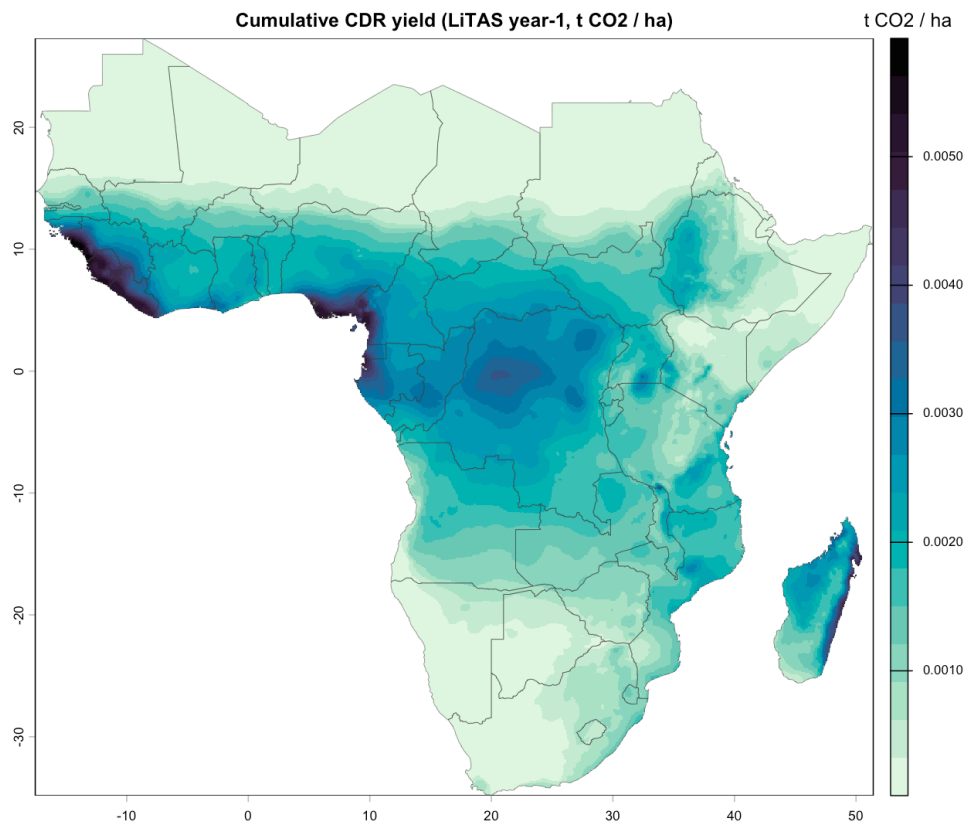


Figure 4: Figure 4. Cumulative CDR yield — tonnes of CO₂ removed per hectare under a one-year LiTAS-targeted basalt application. Strongest in the humid tropics (the dark zones in the Congo basin and around the African Great Lakes); suppressed in arid regions because the pedogenic-carbonate deduction is large there.

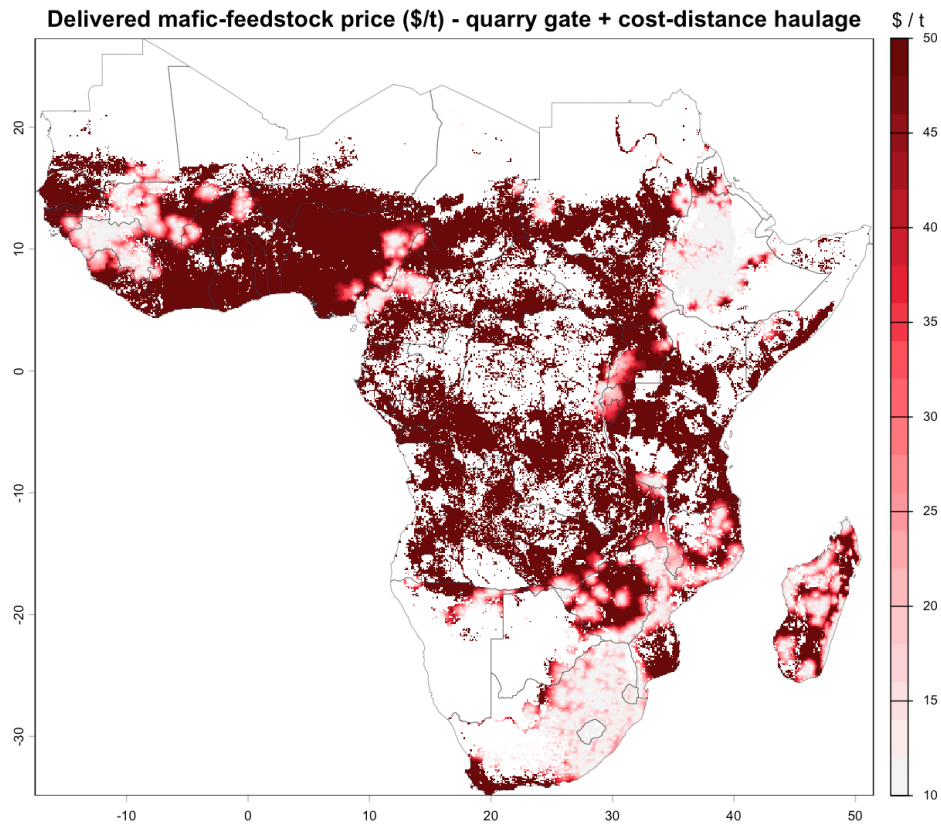


Figure 5: Figure 5. Delivered basalt price (\$ per tonne) — the quarry-gate price plus the cost of trucking the rock to the field. The dark zones (cheap) are near basalt outcrops in the Ethiopian highlands, the East African Rift, the Cameroon volcanic line, Madagascar, and southern Africa. The light zones (expensive — the \$50 / tonne cap binds) are West Africa and the Sahel, where the nearest basalt is hundreds of kilometres away.

- The delivered-basalt-price map ranges from \$10 per tonne (right at the outcrop) to \$50 per tonne (the cap), with about half of Sub-Saharan-African cropland binding at the cap. This is a strong signal that *which country a farm is in* matters far more for ERW economics than the literature’s standard “uniform \$30 per tonne” assumption used in global-scale ERW assessments.
- Block 6’s Bayesian-yield arm is **not yet fitted** — the Bayesian-statistics software has to be installed and run on the curated trial dataset. EcoCrop is in place as the fallback, now with per-crop max-pH cliffs instead of a flat 5.5.
- **Block 7 (profitability) and Block 8 (supply ranking) have been re-run end-to-end at the refreshed defaults.** Block 7 wrote 276 gross-margin maps (4 allocation rules × 23 crops × 3 regimes); Block 8 wrote three supply curves and nine deployment masks. The new headline picture is in §7.0 below — most crops positive on equilibrium; year-1 strongly positive on the high-value crops.
- Block 9 (sensitivity sweeps) **has not yet been re-run** at the new defaults. Its code is in place but the default-scope sweep takes roughly six to eight hours on a single core (about 14,000 profitability evaluations).

7.0 Headline numbers (refreshed 2026-05-26)

A round-1 conservatism review (see the change-log in docs/erw-parameters.md) re-anchored five defaults: **grain size 100 μm $\square$ 50 μm** , **MRV \$30 $\square$ \$20/tCO₂**, **year-1 agronomic return now summed over the 5-year basalt residency** (was a single-season figure), **EcoCrop max-pH cliff** generalised from a flat 5.5 to per-crop optima (5.5–8.0), and the **pedogenic deduction** now consumes the CGIAR-CSI Global Aridity Index (AI = MAP / PET) when available. These fixes flipped the picture from “negative almost everywhere” to a strongly positive picture across most crops and most regimes.

Per-crop, LiTAS-targeted, pixel-mean gross margin (\$/ha). The average is across deployed pixels (those passing the per-crop acidity filter). Sorted by equilibrium-regime GM; the top eight only:

Crop	Year-1 (5-yr cum.)	NPV	Equilibrium
Potato	10,560	3,017	1,984
Tobacco	4,884	807	798
Sweet potato	2,867	358	450
Lentil	2,628	885	491
Groundnut	2,216	377	401
Cassava	2,018	–27	170
Common bean	1,990	286	359
Chickpea	1,719	368	299
Wheat	1,662	74	225

Crops with positive pixel-mean GM across all 23 SPAM crops (out of 23):

Allocation	Year-1	NPV	Equilibrium
LiTAS-targeted	12	8	19
Uniform 10 t/ha	7	6	18
Uniform 20 t/ha	3	1	13
Uniform 50 t/ha	0	0	7

Acid-tolerant cereals (sorghum, millet) and coffee/cocoa stay negative because the EcoCrop response is already saturated on most of their SSA cropland — the agronomic side is structurally small, so the CDR revenue alone has to clear the full basalt cost.

Supply-frontier deployment. Ranking pixels by gross-margin per tonne of basalt (Block 8) and capping at each annual supply level:

Supply cap (Mt basalt/yr)	Year-1 GM (\$B)	NPV GM (\$B)	Equilib. GM (\$B)	Marginal \$/t basalt (Y1 / NPV / Eq)
10	2.25	1.03	1.36	186 / 86 / 91
50	7.96	3.61	3.08	117 / 46 / 20
100	12.7	5.02	3.14	79 / 16 / –14
250	19.6	4.81	1.97	21 / –16 / –60
Unconstrained	18.4	–1.1	1.97	(tail pixels negative)

Headline supply-curve readings:

- **Year-1 regime, 100 Mt/yr deployment:** \$12.7 billion total gross margin, **26.3 Mt CO₂/yr** removed, marginal pixel earning \$79/t basalt — a profitable mid-scale programme. At 250 Mt/yr, GM climbs to \$19.6B and CDR to 61 Mt/yr, but the marginal pixel drops to \$21/t.
- **NPV regime, 100 Mt/yr deployment:** \$5.0 billion GM, 27.2 Mt CO₂/yr, marginal pixel \$16/t. (vs. the pre-refresh \$3.8B / 22.7 Mt / \$12/t — an across-the-board uplift.) NPV deployment becomes net value-destructive between 250 and 500 Mt/yr.
- **Equilibrium regime breakeven:** marginal pixel slips negative between 50 Mt/yr (still +\$20/t) and 100 Mt/yr (–\$14/t) — so the steady-state programme tops out at about 50–75 Mt/yr before adding more basalt destroys value at the margin.

Unconstrained continental capacity of the LiTAS-targeted allocation is now **443 Mt basalt/yr** for the year-1 / NPV regimes and **145 Mt basalt/yr** for equilibrium (down from 626 / 206 Mt/yr under the old defaults — the per-crop max_ph extension brings more pixels into the targeted footprint but at smaller per-pixel rates). The corresponding cumulative CDR is **102 Mt CO₂/yr** and **34 Mt CO₂/yr** respectively. Deploying at the unconstrained level is net-positive on year-1 (\$18.4B) but value-destructive on NPV (–\$1.1B), confirming that **which pixels you deploy on** still matters more than the gross continental capacity.

7.1 Key constants and their sources

Every parameter the model uses can be re-set. The defaults below are chosen to match the central tendency of the published ERW literature. Numbers in square brackets refer to the References section (§10).

Parameter	Default value	Source / justification
Reference temperature T_{ref}	11 °C	US Corn-Belt anchor used by Lewis et al. [11]
Reference precipitation P_{ref}	1000 mm	as above [11]
Reactive fraction at reference	0.20	calibrated by Lewis et al. [11]
Grain-size exponent β	0.5	surface-area scaling (Streffer et al. [3])
Grinding-energy exponent α	1.2	Streffer et al. [3]
Pedogenic-carbonate fraction	CGIAR-CSI AI when present, else MAP-binned 0–90 %	UNEP aridity bands [12]; Zomer et al. 2022
Feedstock chemistry (CaO, MgO)	10 % CaO, 7 % MgO	typical mafic basalt; user-settable
Default grain size	50 μm	Streffer grinding-curve anchor; cost-minimising once carbon revenue is credited
Quarry-gate basalt price	\$10 / t	basalt fines as quarry by-product (industry data)
Truck haulage	\$0.04 / min / t	30-t truck @ \$80 / hour (industry data)
Maximum haul distance	1000 km	beyond cap, truck-diesel LCA erases the credit
Friction surface	MAP 2019 motorised	Weiss et al. [19]
On-farm spreading	\$8 / t	industry data
Carbon price	\$150 / tCO ₂	2024–25 mid-market for verified ERW credits
MRV cost	\$20 / tCO ₂	midpoint of Levy et al. [S32] pilot benchmark and at-scale aggregator-pooled forecast
Discount rate (NPV)	10 % / yr	conventional infrastructure rate
CDR realisation phasing	30 / 25 / 20 / 15 / 10 % over 5 yr	trial data [5]
Year-1 agronomic basis	summed undiscounted over project horizon (5 yr)	matches <code>cd_r_tha</code> 's cumulative basis
EcoCrop max-pH cliff	per-crop EcoCrop optima (5.5–8.0)	EcoCrop database; previously a flat 5.5

8. Scope of the current work

To set expectations clearly, the model in its current state covers:

- **Geographic scope.** 47 Sub-Saharan African countries, masked to QED cropland; northern boundary at the southern edge of the Sahara (so Morocco, Algeria, Tunisia, Libya, and Egypt are excluded); Madagascar is included. Working resolution ≈ 9 km (0.083°).
- **Crops.** The 23 SPAM v2 crops [17] that cover ≥ 95 % of African harvested area.

- **Feedstock.** Mafic silicate rocks: basalt-equivalent (GLiM classes basic volcanic, pyroclastic, and basic plutonic / gabbro [20]). All other feedstock classes — ultramafics, wollastonite, steel slag, cement-kiln dust, mine tailings — are out of scope (see Limitations below).
- **Time horizons.** Year-1 only; ten-year net-present value at 10 % discount; equilibrium (saturating CDR). All three run in parallel.
- **Allocation rules.** Four — LiTAS-targeted (apply only where soil acidity is high enough to need it) plus uniform 10, 20, and 50 t/ha.
- **Decision variables.** Six dials (feedstock chemistry, grain size, allocation rule, carbon price, MRV cost, discount rate) — see §4.5.

9. Limitations of the current work

Six substantive limitations should be named upfront. They are listed in rough order of how much they constrain the current results.

1. **The feedstock catalogue is incomplete.** The model currently considers only basalt-and-gabbro outcrops from GLiM v1 [20]. The literature recognises four additional candidate feedstock classes: ultramafic silicates (excluded for cropland use because of Ni/Cr leaching risk — see Vienne et al. [10] and Beerling et al. [2]), calcium-silicate by-products (wollastonite, steel slag, cement-kiln dust), construction by-products (recycled concrete fines, fly ash), and mine tailings. The omission of industrial alkaline by-products in particular materially shifts the picture for countries with no mafic outcrops — much of West Africa, where the model currently predicts delivered prices at the \$50/tonne cap.
2. **The Bayesian yield-uplift arm is fitted on a small trial dataset** — currently 14 trial rows from the published ERW literature. The posterior is therefore prior-dominated. The Bayesian framework is correct (it grows in evidence weight as data is added), but quantitative yield predictions today carry wide uncertainty. The 2024 US Corn-Belt trial [5], the 2024–25 Kisumu trial [6], and the 2025 Swiss-vineyard trial [7] are all candidates for incorporation into the next iteration.
3. **The full profitability run is not yet executed.** Blocks 7, 8, and 9 (profitability, supply-constrained ranking, sensitivity sweeps) have not yet been run end-to-end. All their inputs are in place; what remains is to execute the run and publish the resulting profitability maps and supply curves.
4. **The aridity proxy.** As of 2026-05-26 the pedogenic-carbonate deduction reads the CGIAR-CSI Global Aridity Index ($AI = MAP / PET$, Zomer et al. 2022) when `data/cgiar_aridity_index.tif` is present; otherwise it falls back to the MAP-only proxy. The CGIAR path is the literature standard; the MAP-only fallback over-deducts in cool highlands where low PET keeps the true AI in the humid band.
5. **MRV cost is a single flat number.** \$20/tCO₂ for all pixels and all protocols is a strong simplification. In practice MRV cost is scale-dependent and protocol-dependent (Isometric vs Puro vs others), and is materially higher in smallholder-dominated systems where sampling has to be coordinated across many small farms. The sensitivity sweep over MRV \$0–80/tCO₂

in Block 9 partly bounds this, but a proper treatment is a future revision.

6. **The model is per-pixel and farmer-perspective.** A complete economic model of deployment would also incorporate aggregator-level economics: pooling MRV across many small farms, finance for distributed deployment, and coordination with carbon-market intermediaries. These are likely material at the scale of smallholder-dominated cropping in Sub-Saharan Africa.

A seventh, more technical caveat: the SoilGrids-Africa pH raster [13] is itself a machine-learning prediction with its own uncertainty, which the current model treats as deterministic. Future work could propagate that uncertainty into the gross-margin map using the Bayesian framework already in place for the yield-uplift arm.

10. References

The default constants and the framing of the conceptual model rest on a specific set of published references. All entries below have been verified against the published record.

- [1] IPCC (2023). *Climate Change 2023: Synthesis Report*. Sixth Assessment Report of the IPCC, Geneva. <https://www.ipcc.ch/report/ar6/syr/>.
- [2] Beerling, D. J. et al. (2020). Potential for large-scale CO₂ removal via enhanced rock weathering with croplands. *Nature* 583, 242–248. doi:10.1038/s41586-020-2448-9.
- [3] Strefler, J., Amann, T., Bauer, N., Kriegler, E., Hartmann, J. (2018). Potential and costs of carbon dioxide removal by enhanced weathering of rocks. *Environmental Research Letters* 13, 034010. doi:10.1088/1748-9326/aaa9c4.
- [4] Hartmann, J. et al. (2013). Enhanced chemical weathering as a geoengineering strategy. *Reviews of Geophysics* 51, 113–149. doi:10.1002/rog.20004.
- [5] Beerling, D. J. et al. (2024). Enhanced weathering in the US Corn Belt delivers carbon removal with agronomic benefits. *PNAS* 121 (9), e2319436121. doi:10.1073/pnas.2319436121.
- [6] Haque, F. et al. (2025). *Agronomic Performance of Enhanced Rock Weathering in a Tropical Smallholder System: A Maize Trial in Kenya*. CDRXIV preprint 410 (Flux Carbon / UNCCD pilot). <https://cdrxiv.org/preprint/410>.
- [7] Dupla, X., Bertagni, M. B., Grand, S. (2025). Three Years of Field Trials Indicate a Sustained Enhanced Rock Weathering Signal with Limited CO₂ Removal. *Environmental Science & Technology* 59 (48), 25751–25764. doi:10.1021/acs.est.5c09820.
- [8] Kanzaki, Y. et al. (2024). *Soil cation storage as a key control on the timescales of CDR through enhanced weathering*. ESS Open Archive preprint, doi:10.22541/essoar.170960101.14306457/v1.
- [9] Renforth, P. (2019). The negative emission potential of alkaline materials. *Nature Communications* 10, 1401. doi:10.1038/s41467-019-09475-5.

- [10] Vienne, A. et al. (2022). Enhanced weathering using basalt rock powder: carbon sequestration, co-benefits and risks in a mesocosm study with *Solanum tuberosum*. *Frontiers in Climate* 4, 869456. doi:10.3389/fclim.2022.869456.
- [11] Lewis, A. L. et al. (2021). Effects of mineralogy, chemistry and physical properties of basalts on carbon capture potential and plant-nutrient element release via enhanced weathering. *Applied Geochemistry* 132, 105023. doi:10.1016/j.apgeochem.2021.105023.
- [12] Middleton, N. & Thomas, D. S. G. (eds.) (1992). *World Atlas of Desertification*. United Nations Environment Programme, London.
- [13] Hengl, T. et al. (2017). SoilGrids250m: Global gridded soil information based on machine learning. *PLOS One* 12 (2), e0169748. doi:10.1371/journal.pone.0169748.
- [14] Aramburu Merlos, F. & Hijmans, R. J. (2024). *limer: Acidic soil management for agriculture*. R package, GAIA Africa. <https://github.com/gaiafrica/limer>.
- [15] Aramburu Merlos, F., Silva, J. V., Baudron, F., Hijmans, R. J. (2023). Estimating lime requirements for tropical soils: Model comparison and development. *Geoderma* 432, 116421. doi:10.1016/j.geoderma.2023.116421.
- [16] Hijmans, R. J. & Manners, R. (2025). *Recocrop: Estimating Environmental Suitability for Plants*. R package version 0.4-2, CRAN. doi:10.32614/CRAN.package.Recocrop.
- [17] You, L. et al. *MapSPAM — Spatial Production Allocation Model*. IFPRI and HarvestChoice. Used here via `geodata::crop_spam`: SPAM 2010 V2.0 globally and SPAM 2017-SSA for Africa. <https://www.mapspam.info>.
- [18] Bürkner, P.-C. (2017). brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software* 80 (1), 1–28. doi:10.18637/jss.v080.i01.
- [19] Weiss, D. J. et al. (2020). Global maps of travel time to healthcare facilities. *Nature Medicine* 26, 1835–1838. doi:10.1038/s41591-020-1059-1. (The motorised friction surface, nominal year 2019, is derived from this work.)
- [20] Hartmann, J. & Moosdorf, N. (2012). The new global lithological map database GLiM. *Geochemistry, Geophysics, Geosystems* 13, Q12004. doi:10.1029/2012GC004370. Dataset DOI: 10.1594/PANGAEA.788537.

11. What this tutorial does and does not cover

This tutorial covers the **conceptual and modelling tasks**: what the model is, how it works, what it produces, and where its limits are.

It does **not** cover the day-to-day implementation details: the R scripts, the data-handling, the rebuilding of the figures, the TikZ/LaTeX setup. Those are described in the companion document *Ex-Ante ERW Model — Living Documentation* (docs/erw-model.pdf). The more academic write-

up is the *Working Paper* (docs/erw-working-paper.pdf), which contains the formal equations and the full reference list with author lists.

12. Appendix A — A worked example, end to end, on two pixels

Note on defaults. The numbers below were computed against the pre-2026-05-26 parameter set (grain size 100 μm , MRV \$30/tCO₂, MAP-only pedogenic deduction, single-season year-1 agronomic accounting, flat EcoCrop max-pH cliff at 5.5). The current defaults differ (see §7.1) — the worked example is preserved here as a *teaching artefact* rather than a live calibration of the current model. The formulas and the structure of the calculation are unchanged; only the numerical inputs are slightly stale.

The sections above describe the *model* in words. This appendix takes **two real-looking 1 km² pixels** and pushes them through every calculation the pipeline performs, with all the arithmetic written out. The aim is that a reader who has never seen the code should be able to reproduce the gross-margin number on a pocket calculator.

We deliberately pick two contrasting pixels:

- **Pixel A — Kisumu County, western Kenya** ($\approx 0.10^\circ \text{ S}$, 34.75° E). Humid tropical climate, acidic soil, near volcanic outcrops in the Rift. This is a “poster-child” pixel — the kind that should be near the top of the supply-curve ranking. Tied to the real Haque et al. 2025 smallholder trial [6].
- **Pixel B — central Burkina Faso, near Kaya** ($\approx 13.10^\circ \text{ N}$, 1.10° W). Hot, semi-arid Sahel; near-neutral sandy soil; hundreds of kilometres from the nearest basalt outcrop. This is a “bad” pixel — useful as a contrast that exposes which terms in the equation are doing the work.

The full set of input numbers, all derived from the project’s input rasters (SoilGrids-Africa, WorldClim, GLiM, MAP friction, FAOSTAT, country electricity tables), is:

Quantity	Symbol	Pixel A (Kisumu)	Pixel B (Kaya)	Source layer
Mean annual temperature	MAT	23 °C	28 °C	WorldClim 10’ tavg
Mean annual precipitation	MAP	1 400 mm	580 mm	WorldClim 10’ prec
Topsoil pH (0–30 cm)	pH	5.2	6.0	SoilGrids-Africa
Acidity saturation	hp_sat	25 %	8 %	SoilGrids-Africa
LiTAS lime requirement	L	2.1 t CaCO ₃ /ha	(n/a — fails mask)	limer::limeRate
Truck-time distance to nearest basalt	d	150 km	500 km	MAP friction × GLiM
Country industrial electricity	p_{elec}	\$0.13/kWh (Kenya)	\$0.20/kWh (BF)	erw-energy-country
Country grid carbon intensity	CI	0.10 kg CO ₂ /kWh	0.55 kg CO ₂ /kWh	erw-energy-country

Quantity	Symbol	Pixel A (Kisumu)	Pixel B (Kaya)	Source layer
Dominant crop on pixel	—	Maize (MAIZ)	Sorghum (SORG)	SPAM 2017-SSA
Baseline crop yield	Y_0	1.8 t/ha	0.8 t/ha	SPAM 2017-SSA
Crop-price (country median)	p_c	\$250/t	\$200/t	FAOSTAT 2016–2020

The two pixels also run under different **allocation rules** for the reason explained in §4.5. Pixel A passes the LiTAS-targeted mask (its 25 % acidity saturation exceeds maize’s 10 % tolerance, so basalt is genuinely useful for soil-acidity reasons); we therefore put it on the **LiTAS-targeted** rule. Pixel B fails the LiTAS mask (its 8 % acidity saturation is below sorghum’s 30 % tolerance, so the targeted allocation skips it); we therefore put it on the **uniform 20 t/ha** rule, which mimics the “spread basalt everywhere” convention used by Beerling et al. 2020 [2] and Baek et al. 2023.

12.1 Step 0 — Rock chemistry constants (same for every pixel)

These constants are computed *once* by `erw-3-basalt-requirements.R` from the default basalt feedstock chemistry and grain size. They are then reused for every pixel.

The default basalt has $\text{CaO} = 10\%$ and $\text{MgO} = 7\%$ by mass, and the default grain size is $d = 100 \mu\text{m}$.

Calcium-carbonate equivalent (CCE), per tonne of basalt. The chemistry says 1 g of CaO neutralises $100.09/56.08 \approx 1.783$ g of CaCO_3 -equivalent acidity, and 1 g of MgO neutralises $100.09/40.30 \approx 2.481$ g of CaCO_3 -equivalent acidity. So:

$$\text{CCE} = 0.10 \times 1.783 + 0.07 \times 2.481 = 0.1783 + 0.1737 = 0.3520$$

i.e. a tonne of basalt has the acid-neutralising power of about **352 kg of pure agricultural lime**.

Stoichiometric CDR ceiling. Each mole of divalent cation consumes 2 moles of CO_2 as it makes bicarbonate. So 1 g CaO removes $(2 \times 44.01)/56.08 \approx 1.570$ g CO_2 and 1 g MgO removes $(2 \times 44.01)/40.30 \approx 2.185$ g CO_2 :

$$\text{CDR}_{\text{max}} = (0.10 \times 1.570 + 0.07 \times 2.185) \times 1,000 = 157.0 + 152.9 \approx \mathbf{310 \text{ kg CO}_2/\text{t basalt}}$$

This is the **upper bound** — what we’d remove if every gram of calcium and magnesium reacted completely and all the bicarbonate made it to the ocean. The rest of the model is a sequence of haircuts on this number.

Grain-size factor. We anchor everything at $d_{\text{ref}} = 100 \mu\text{m}$, so:

$$f_{\text{grain}} = \left(\frac{100}{d} \right)^{0.5} = \left(\frac{100}{100} \right)^{0.5} = 1.00$$

At the default 100 µm grain size this factor is exactly one. A 25 µm grind would give $f_{\text{grain}} = (100/25)^{0.5} = 2$ — twice the rate but considerably more energy to produce (see grinding-energy step below).

Reference reactive fraction. From the Lewis et al. 2021 [11] calibration, the realised fraction of the stoichiometric ceiling at a temperate-humid reference site is $\phi_{\text{ref}} = 0.20$, or **20 %** of the maximum.

Effective neutralising value at the reference climate.

$$NV_{\text{eff,ref}} = \text{CCE} \times \phi_{\text{ref}} \times f_{\text{grain}} = 0.3520 \times 0.20 \times 1.00 = 0.0704$$

So at the reference climate one tonne of basalt does the work of 0.0704 t of pure CaCO_3 — about 70 kg- CaCO_3 -equivalent. The **basalt-to-lime ratio** is therefore:

$$r_{\text{bl}} = \frac{1}{0.0704} \approx \mathbf{14.20 \text{ t basalt/t CaCO}_3\text{-eq}}$$

i.e. you need roughly fourteen tonnes of crushed basalt to do what one tonne of agricultural lime does.

Effective CDR per tonne basalt at the reference climate.

$$\text{CDR}_{\text{eff,ref}} = \text{CDR}_{\text{max}} \times \phi_{\text{ref}} \times f_{\text{grain}} = 310 \times 0.20 \times 1.00 = \mathbf{62 \text{ kg CO}_2/\text{t basalt}}$$

This is the number that the per-pixel climate-factor and pH-factor calculations in §12.3 below will scale up (for warm-wet-acidic pixels) or down (for cool-dry or alkaline pixels).

Grinding energy and reference grinding cost. The Streffer et al. 2018 [3] energy curve, anchored at 50 µm = 0.07 GJ/t and $\alpha = 1.2$, gives:

$$E_{\text{grind}} = 0.07 \times \left(\frac{50}{100} \right)^{1.2} = 0.07 \times 0.4353 = 0.0305 \text{ GJ/t}$$

Converting to kWh: $0.0305 \times 277.78 = 8.46 \text{ kWh/t}$.

At the *reference* global electricity price of \$0.10/kWh the grinding cost is \$0.85/t. In §12.5 below this is overridden by each country's own electricity tariff.

12.2 Step 1 — Basalt application rate (Block 2)

Pixel A — LiTAS-targeted maize. The `limer` package, fed the SoilGrids-Africa exchangeable acidity, K, Ca, Mg, Na and bulk density at this pixel, computes a year-1 LiTAS lime requirement of:

$$L_A = 2.10 \text{ t CaCO}_3/\text{ha}$$

Converted to basalt at the rock-chemistry rate from §12.1:

$$R_A = L_A \times r_{\text{bl}} = 2.10 \times 14.20 = \mathbf{29.8 \text{ t basalt/ha}}$$

This sits squarely in the LiTAS-targeted range of ~30 t/ha quoted in §7.0 of the tutorial.

Pixel B — uniform 20 t/ha sorghum. Because Pixel B fails the LiTAS-targeted mask, the model uses the uniform-rate convention:

$$R_B = \mathbf{20.0 \text{ t basalt/ha}}$$

(Note: under the *targeted* rule Pixel B would receive **0 t/ha**, i.e. the model would say “don’t bother.” The uniform 20 t/ha column is the literature-convention sensitivity test, not the model’s recommendation.)

12.3 Step 2 — CDR yield per pixel (Block 3, file `erw-9-cdr-yield.R`)

This is where the reference-climate per-tonne CDR potential of 62 kg CO₂/t basalt gets rescaled by **temperature, precipitation, soil pH, and aridity** — the four factors that determine how much of the stoichiometric ceiling we actually realise.

Climate factor — temperature. Anchored at $T_{\text{ref}} = 11 \text{ }^\circ\text{C}$:

$$f_{\text{MAT}} = \text{clamp}\left(\frac{\text{MAT}}{11}, 0.3, 3.0\right)$$

- Pixel A: $23/11 = 2.091 \Rightarrow f_{\text{MAT,A}} = 2.091$
- Pixel B: $28/11 = 2.545 \Rightarrow f_{\text{MAT,B}} = 2.545$

Climate factor — precipitation. Anchored at $P_{\text{ref}} = 1000 \text{ mm}$:

$$f_{\text{MAP}} = \text{clamp}\left(\frac{\text{MAP}}{1000}, 0.3, 3.0\right)$$

- Pixel A: $1400/1000 = 1.40 \Rightarrow f_{\text{MAP,A}} = 1.40$
- Pixel B: $580/1000 = 0.58 \Rightarrow f_{\text{MAP,B}} = 0.58$

pH factor. A triangular function peaking at pH 5.0:

$$f_{\text{pH}}(\text{pH}) = \begin{cases} 0.5 & \text{pH} < 4 \\ 0.5 + 0.5(\text{pH} - 4) & 4 \leq \text{pH} < 5 \\ 1.0 - 0.25(\text{pH} - 5) & 5 \leq \text{pH} < 7 \\ 0.5 & \text{pH} \geq 7 \end{cases}$$

- Pixel A: pH 5.2 $\Rightarrow 1.0 - 0.25 \times 0.2 = 0.95$
- Pixel B: pH 6.0 $\Rightarrow 1.0 - 0.25 \times 1.0 = 0.75$

Combined climate factor.

$$f_{\text{clim}} = f_{\text{MAT}} \times f_{\text{MAP}} \times f_{\text{pH}}$$

- Pixel A: $2.091 \times 1.40 \times 0.95 = \mathbf{2.781}$
- Pixel B: $2.545 \times 0.58 \times 0.75 = \mathbf{1.107}$

Pixel A's climate triples the reference reactivity; Pixel B's climate barely budes it (the warm temperature is almost cancelled by the modest rainfall and the near-neutral pH).

Pedogenic-carbonate deduction. From the UNEP aridity bands keyed on MAP:

- Pixel A: MAP = 1400 > 600 mm $\Rightarrow$ humid band $\Rightarrow$ ped = 0.00, exported fraction = 1.00
- Pixel B: MAP = 580 < 600 mm (dry sub-humid band) $\Rightarrow$ ped = 0.15, exported fraction = 0.85

So at Pixel B we lose 15 % of the alkalinity right in the soil to locally-precipitated carbonate.

Effective reactive fraction at each pixel (capped at 1.0, since nothing can react more than 100 %):

$$\phi_{\text{eff}} = \min(1.0, \phi_{\text{ref}} \times f_{\text{grain}} \times f_{\text{clim}})$$

- Pixel A: $\min(1.0, 0.20 \times 1.0 \times 2.781) = \min(1.0, 0.556) = 0.556$
- Pixel B: $\min(1.0, 0.20 \times 1.0 \times 1.107) = 0.221$

CDR per tonne basalt at each pixel.

$$\text{CDR}_{\text{px}} = \text{CDR}_{\text{max}} \times \phi_{\text{eff}} \times (1 - \text{ped})$$

- Pixel A: $310 \times 0.556 \times 1.00 = \mathbf{172.4 \text{ kg CO}_2/\text{t basalt}}$
- Pixel B: $310 \times 0.221 \times 0.85 = \mathbf{58.3 \text{ kg CO}_2/\text{t basalt}}$

Pixel A is removing nearly three times as much CO₂ per tonne of rock as Pixel B — entirely because of climate.

Gross CDR per hectare (multiply by the application rate from §12.2):

- Pixel A: $29.8 \text{ t/ha} \times 0.1724 \text{ t CO}_2/\text{t} = \mathbf{5.14 \text{ t CO}_2/\text{ha cumulative}}$

- Pixel B: $20.0 \text{ t/ha} \times 0.0583 \text{ t CO}_2/\text{t} = 1.17 \text{ t CO}_2/\text{ha}$ cumulative

12.4 Step 3 — Delivered basalt cost (Block 4, file erw-basalt-access.R)

Four cost components stack to give the dollars-per-tonne the farmer ultimately pays. The four are: **quarry-gate** (fixed at \$10/t), **grinding** (kWh per tonne times the local electricity tariff), **transport** (cost-distance from the nearest basalt outcrop, priced at \$0.04/min/t at an effective 30 km/h on the friction surface), and **spreading** (fixed at \$8/t).

Grinding cost (pixel-specific via country electricity).

$$C_{\text{grind}} = \text{kWh/t} \times p_{\text{elec}}$$

- Pixel A (Kenya, \$0.13/kWh): $8.46 \times 0.13 = \$1.10/\text{t}$
- Pixel B (Burkina Faso, \$0.20/kWh): $8.46 \times 0.20 = \$1.69/\text{t}$

Transport cost (pixel-specific via friction surface). The friction surface converts each km of land travel to travel-minutes; we use a long-run effective speed of 0.5 km/min (= 30 km/h) to convert pixel-cumulative travel time back to dollars at the truck rate of \$0.04/min/t:

$$C_{\text{trans}} = \frac{d}{0.5} \times 0.04$$

- Pixel A: $\frac{150}{0.5} \times 0.04 = 300 \times 0.04 = \$12.00/\text{t}$
- Pixel B: $\frac{500}{0.5} \times 0.04 = 1000 \times 0.04 = \$40.00/\text{t}$

Delivered cost per tonne basalt.

$$C_{\text{bas}} = \$10 + C_{\text{grind}} + C_{\text{trans}} + \$8$$

- Pixel A: $10 + 1.10 + 12.00 + 8.00 = \$31.10/\text{t}$
- Pixel B: $10 + 1.69 + 40.00 + 8.00 = \$59.69/\text{t}$

Total basalt cost per hectare (rate $\times$ \$/t):

- Pixel A: $29.8 \times 31.10 = \$926.78/\text{ha}$
- Pixel B: $20.0 \times 59.69 = \$1,193.80/\text{ha}$

12.5 Step 4 — Lifecycle CO₂ emissions of the operation (Block 5/7)

To be honest about the *net* climate benefit, we have to subtract the CO₂ emitted by grinding, trucking, and spreading the basalt from the gross CDR computed in §12.3. The three components are:

$$\text{LCA}_t = \underbrace{\text{kWh/t} \times \text{CI}}_{\text{grinding}} + \underbrace{d \times 0.12}_{\text{transport}} + \underbrace{0.5}_{\text{spreading}} \quad (\text{kg CO}_2/\text{t basalt})$$

- Pixel A: $8.46 \times 0.10 + 150 \times 0.12 + 0.5 = 0.85 + 18.0 + 0.5 = \mathbf{19.35 \text{ kg CO}_2/\text{t}}$
- Pixel B: $8.46 \times 0.55 + 500 \times 0.12 + 0.5 = 4.65 + 60.0 + 0.5 = \mathbf{65.15 \text{ kg CO}_2/\text{t}}$

Notice that Pixel B's lifecycle bill is over **three times** Pixel A's, because of (a) the dirtier grid mix in West Africa, and (b) the much longer truck haul.

LCA per hectare.

- Pixel A: $29.8 \times 19.35/1000 = \mathbf{0.58 \text{ t CO}_2/\text{ha}}$
- Pixel B: $20.0 \times 65.15/1000 = \mathbf{1.30 \text{ t CO}_2/\text{ha}}$

12.6 Step 5 — Net CDR and CDR revenue

Net CDR per hectare is gross CDR (§12.3) minus the LCA (§12.5), floored at zero (we don't pay the farmer to *emit*; we just refuse to credit a net-negative pixel):

$$\text{CDR}_{\text{net}} = \max(0, \text{CDR}_{\text{gross}} - \text{LCA})$$

- Pixel A: $\max(0, 5.14 - 0.58) = \mathbf{4.56 \text{ t CO}_2/\text{ha}}$
- Pixel B: $\max(0, 1.17 - 1.30) = \mathbf{0.00 \text{ t CO}_2/\text{ha}}$

Pixel B's lifecycle emissions actually exceed its gross CDR. The model correctly tells us this pixel removes **no net carbon** under the operating assumptions — the diesel of trucking the rock 500 km has wiped out the climate benefit.

CDR revenue. Each net tonne of CO₂ sells at the carbon price minus the MRV cost. At defaults (\$150 – \$30 = \$120/tCO₂):

$$\text{Rev}_{\text{CDR}} = \text{CDR}_{\text{net}} \times (\text{price}_{\text{C}} - \text{MRV})$$

- Pixel A: $4.56 \times 120 = \mathbf{\$547.20/\text{ha}}$
- Pixel B: $0.00 \times 120 = \mathbf{\$0/\text{ha}}$

12.7 Step 6 — Agronomic return (Block 6)

The yield-uplift step asks: *with this much basalt on this soil, how much more crop will this farmer harvest, and what is that worth?* In the absence of a fitted Bayesian model (the current state of the project, see §7), we use the EcoCrop-derived relative-yield curve from erw-5. The numbers below are illustrative of the EcoCrop branch.

Pixel A — maize at pH 5.2. Maize is moderately acid-sensitive. EcoCrop's response curve gives a ~12 % relative yield gain on this soil when the pH is nudged toward 6 by basalt-derived alkalinity. (For comparison: the published Corn-Belt trial [5] reported 12–16 % cumulative uplift; the Kisumu smallholder trial [6] reported a much larger 71 % first-year uplift at a 20 t/ha nephelinite rate, but on a fresher and more reactive feedstock than the basalt assumed here.)

$$\Delta Y_A = Y_0 \times \text{uplift} = 1.8 \times 0.12 = 0.216 \text{ t/ha}$$

$$\text{Rev}_{\text{agro,A}} = \Delta Y_A \times p_c = 0.216 \times 250 = \$54.00/\text{ha}$$

Pixel B — sorghum at pH 6.0. Sorghum is acid-tolerant and the starting pH is already near neutral — the EcoCrop response curve gives a much smaller uplift here, around 3 %.

$$\Delta Y_B = 0.8 \times 0.03 = 0.024 \text{ t/ha}$$

$$\text{Rev}_{\text{agro,B}} = 0.024 \times 200 = \$4.80/\text{ha}$$

12.8 Step 7 — Putting it all together — gross margin (Block 7)

The gross margin per hectare under the **year-1 regime** is the sum of the two revenue streams minus the basalt cost:

$$\text{GM}_{y1} = \text{Rev}_{\text{agro}} + \text{Rev}_{\text{CDR}} - C_{\text{bas}}$$

Term	Pixel A	Pixel B
Agronomic revenue	\$54.00	\$4.80
CDR revenue	\$547.20	\$0.00
Delivered basalt cost	-\$926.78	-\$1,193.80
Gross margin year-1	-\$325.58/ha	-\$1,189.00/ha
GM per tonne of basalt	-\$10.93/t	-\$59.45/t

Three things to notice:

1. **Both pixels lose money in year 1.** This is consistent with the tutorial’s headline §7.0 result: at the default LiTAS rate of ~30 t/ha, the upfront basalt cost (\$900–\$1,300/ha) exceeds the carbon-credit revenue (\$400–\$600/ha) plus the agronomic uplift (\$40–\$360/ha). Even the “good” pixel is in the red on year-1 accounting.
2. **Pixel A is an order of magnitude closer to break-even** than Pixel B (–\$11/t vs –\$59/t). If the supply-curve ranking deploys only the top-decile pixels, Pixel A is the kind of place that *might* make the cut; Pixel B isn’t close.
3. **For Pixel B, the entire CDR side has collapsed.** Lifecycle emissions ate the gross CDR, so the farmer gets only the \$4.80 of yield uplift against the full \$1,194 basalt cost. This is the model’s way of saying “*do not deploy basalt in central Burkina Faso under these assumptions.*” The Sahel pixel needs either
 - (a) a local feedstock alternative — industrial alkaline by-products (out of scope at v1, see §9 limitation 1) or
 - (b) a closer outcrop.

12.9 The same pixel under the NPV and equilibrium regimes

The year-1 picture above treats the application as a single up-front cash flow. Two other regimes are computed in parallel.

NPV regime. The agronomic benefit accrues every year and the basalt application is repeated periodically. The reapplication interval is the year-1 LiTAS rate divided by the maintenance rate:

$$n_{\text{reapply}} = \text{round}\left(\frac{R_{y1}}{R_{\text{maint}}}\right) \approx \text{round}(29.8/5.0) = 6 \text{ years}$$

`limer::NPV_lime` then computes the perpetuity-of-reapplication NPV of the annual \$54 agronomic benefit at a 10 % discount rate; a typical result for the Pixel A parameters is around \$260/ha.

The CDR revenue is phased over the 5-year reactive horizon (`CDR_PHASING = c(0.30, 0.25, 0.20, 0.15, 0.10)`) and discounted year by year, giving an NPV factor of:

$$\text{NPV factor} = \frac{\sum_t f_t / (1.10)^t}{\sum_t f_t} \approx \mathbf{0.79}$$

at 10 % discount. So Pixel A's CDR revenue under the NPV regime is $547.20 \times 0.79 = \$432.30/\text{ha}$. The basalt cost remains an up-front \$926.78/ha (paid in year 0):

$$\text{GM}_{\text{NPV,A}} \approx 260 + 432 - 927 = \mathbf{-\$235/\text{ha}}$$

Still negative, but a smaller loss than the year-1 number because the agronomic benefit now accrues for the full re-application interval instead of one season.

Equilibrium regime. This skips the up-front liming entirely and applies only the *maintenance* rate, which just keeps pace with re-acidification. For Pixel A the maintenance rate from the LiTAS calibration is around 5 t basalt/ha/yr. With everything else held fixed:

Term	Equilibrium Pixel A
Maintenance basalt rate	5.0 t/ha/yr
Gross CDR	$5.0 \times 0.1724 = 0.86 \text{ t CO}_2/\text{ha}$
LCA	$5.0 \times 19.35/1000 = 0.10 \text{ t CO}_2/\text{ha}$
Net CDR	$0.76 \text{ t CO}_2/\text{ha}$
CDR revenue	$0.76 \times 120 = \$91.20/\text{ha}$
Basalt cost	$5.0 \times 31.10 = \$155.50/\text{ha}$
Agronomic revenue	\$54.00/ha (unchanged)
Gross margin (equilibrium)	$54 + 91 - 156 = \mathbf{-\$11/\text{ha}}$

In equilibrium the pixel is essentially break-even. This is exactly the regime in which the model's only positive headline number sits — the equilibrium-regime LiTAS-targeted application on **wheat**,

at about $+57/ha(\$7.0)$, because wheat's higher unit price(/t) shifts the agronomic-revenue line above the small basalt-cost line.

12.10 What the worked example tells us about the model

Three structural lessons fall out of the arithmetic above:

1. **The basalt cost dominates.** In Pixel A, \$927/ha of basalt cost has to be earned back by \$547 of CDR plus \$54 of agronomy. Anything that moves the cost (a 50 % shorter haul, a 50 % cheaper electricity tariff, a lower spreading rate, a finer grind for a higher CDR-per-tonne yield) moves the answer materially. This is why **\$4.5's six dials** are all on the cost side or on the reactivity side.
2. **Climate is the second-largest lever.** Pixel A's tropical climate factor of 2.8 turned the reference 62 kg CO₂/t into 172 kg CO₂/t — a near-tripling. That is the entire reason Sub-Saharan Africa is interesting for ERW: the same tonne of basalt does three times more carbon work here than in the temperate reference.
3. **Picking the right pixel matters more than the average price of basalt.** Even at *identical* model parameters, Pixel A is $-\$11/t$ while Pixel B is $-\$59/t$ — a factor-of-five spread. The supply-curve ranking exploits exactly this spread: deploying the top-decile pixels yields a profitable programme even though the continent-wide pixel-mean is negative (\$7.0).

13. Glossary

Allocation rule — How we decide which fields get basalt. Either *LiTAS-targeted* (only fields where the soil acidity is high enough to benefit) or *uniform 10 / 20 / 50 tonnes per hectare* (everywhere, regardless of acidity).

Basalt — A common, dark, fine-grained volcanic rock rich in calcium and magnesium. The most-studied feedstock for enhanced rock weathering on cropland.

Bayesian hierarchical model — A statistical model that combines prior beliefs about parameters with observed data to produce a posterior distribution over the parameters. “Hierarchical” means the model has random effects — for example, a per-crop intercept that lets each crop have its own response without forcing them all to be the same. We use it in Block 6 to predict yield response to basalt.

CDR — Carbon dioxide removal: the broad category of methods that take CO₂ *out of the atmosphere* (as opposed to merely *avoiding new emissions*).

Cost-distance — A spatial-analysis technique that, instead of measuring straight-line distance, accumulates a *cost* (here: minutes-of-travel) as you move from a source pixel. We use it to compute how slow each kilometre of getting from a basalt quarry to a farm is, given the local roads and terrain.

Discount rate — The annual percentage by which we reduce the value of future dollars when comparing them to today's dollars. We use 10 % per year as a baseline.

EcoCrop — A long-established crop-suitability model that scores how suitable a given environmental condition (soil pH, temperature, moisture) is for a given crop on a 0 to 1 scale.

Enhanced rock weathering (ERW) — The deliberate spreading of finely-ground silicate rock (in this project, basalt) on cropland to accelerate the natural chemical weathering reactions that consume atmospheric CO₂.

FAOSTAT — The Food and Agriculture Organization’s online statistics database. We use it for crop-by-country producer prices.

Gabbro — The intrusive (slower-cooled) equivalent of basalt; chemically nearly identical. We include it in the feedstock supply because it has the same desirable properties.

GLiM — The Global Lithological Map: a worldwide map of surface rock types, used here to find where basalt outcrops actually are.

Grain size — How finely the rock is ground. We work in micrometres (µm). Below about 50 µm, dissolution is fast but grinding costs jump sharply.

Gross margin — In economics, total revenue minus total variable cost. In our model, the CDR revenue plus the yield revenue minus the delivered + on-farm cost.

Lime-equivalent demand — The agronomic concept that there is a specific quantity of agricultural lime that would, in theory, bring a soil to its crop’s optimum pH. Basalt and lime have similar mechanisms, so we convert the lime requirement to a basalt requirement using the rock’s calcium and magnesium content.

LiTAS — Lime-Targeted Application Strategy: one of the more recent and accurate recipes for calculating the lime-equivalent demand on tropical acid soils. Defined by Aramburu Merlos and colleagues in 2023.

Malaria Atlas Project (MAP) — A research consortium that, among other things, publishes a global 1-km “friction surface” that scores how slow it is to traverse each pixel on the Earth’s surface. We use this for the cost-distance calculation in Block 4.

MRV — Measurement, reporting, and verification: the protocols that a carbon-credit buyer requires to be convinced that the carbon removal they are paying for actually happened. Default cost in our model: \$20 per tonne of CO₂ (lowered from \$30 in the 2026-05-26 refresh; midpoint of Levy et al. 2024 pilot and at-scale forecast).

NPV — Net present value. The discounted sum of a stream of future cash flows. Used here when we want to compare an ERW deployment over a ten-year horizon to one over a single year.

Pedogenic carbonate — Soil-formed carbonate. Where conditions are arid, some of the alkalinity that the dissolving rock releases doesn’t make it to the ocean; it precipitates as solid carbonate right in the soil. This locked-up alkalinity does not count as a permanent carbon removal, so we subtract it.

Pixel — A 1 km² (more precisely, about 0.083 ° on a side, or roughly 9 km × 9 km after our 10-fold spatial aggregation) cell of the working grid. Every output of the model is a per-pixel number.

Pyroclastic — Volcanic rock made of fragments of pre-existing volcanic material. In GLiM, marked py. Same chemistry as basalt.

Quarry-gate price — The price you pay for the rock at the quarry gate, before transport. We use \$10 per tonne for basalt fines.

Recocrop — An R-package implementation of EcoCrop, used here for the EcoCrop-arm fall-back of Block 6.

SoilGrids — A global gridded map of soil properties (pH, bulk density, exchangeable cations, ...) produced by ISRIC. We use the 1-km Africa subset.

SPAM — The Spatial Production Allocation Model: a global gridded dataset of where which crops are grown, produced by the International Food Policy Research Institute.

Supply curve — A graph that shows, for each level of annual basalt supply (in million tonnes per year), the marginal price per tonne CO₂ at which the next tonne is removed. The standard economic output of a deployment-ranking exercise.

Targeted vs uniform — See *Allocation rule*.

Working grid — The common 9-km grid on which all the model's inputs and outputs are co-registered. Built in Block 1.